IVAN ANICH

EVERYTHING I KNOW ABOUT TIME SERIES

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Difference Equations

This chapter is essentially a distillation of Hamilton's [1] chapter of the same name.

[1] James D Hamilton. *Time series analysis*, volume 2. Princeton Princeton University Press, 1994

The whole point of time series analysis is to uncover and describe the statistical properties and structure of a process across time. Certain processes may have a lot of structure, so much that advanced non-linear methods such as those enabled by machine learning are needed. Others have very little, and must be described by a very simple model as we are always worried about overfitting. So, we begin by studying the simplest of processes.

Assume we have the deterministic process

$$y_t = \phi y_{t-1} + w_t \tag{1}$$

In order to find the state of this process at any time t, we need some kind of initial condition. We may take y_{-1} as given, or if we want to look some arbitrary time setp j into the future from a given time t, we may take y_{t-1} as given. In either case, we can use recursion to find the state of y_t at any future time t:

$$\begin{aligned}
y_t &= \phi y_{t-1} + w_t, \\
y_{t+1} &= \phi y_t + w_{t+1} && = \phi(\phi y_{t-1} + w_t) + w_{t+1} && = \phi^2 y_{t-1} + \phi w_t + w_{t+1}, \\
y_{t+2} &= \phi y_{t+1} + w_{t+2} && = \phi(\phi^2 y_{t-1} + \phi w_t + w_{t+1}) + w_{t+2} && = \phi^3 y_{t-1} + \phi^2 w_t + \phi w_{t+1} + w_{t+2}, \\
&\vdots \\
y_{t+j} &= \phi^{j+1} y_{t-1} + \sum_{k=0}^{j} \phi^{j-k} w_{t+k}.
\end{aligned} \tag{2}$$

The last line shows the general form of this process at some future state: it has the classic characteristic of a time series process in that its state at any point in time depends on two things: an initial condition y_{t-1} that encapsulates all of its history up until that point, and the shocks w_{t+k} that have affected the process since the initial condition. The effect of a shock at time t on the system at some future time $t + j$ is

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^j, \tag{3}$$

and the effect of the initial condition is

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^{j+1}. \tag{4}$$

So, it is clear that the stability of the process depends on ϕ. If $-1 < \phi < 1$, the effect of the initial condition and all shocks at some future horizon j goes to zero; if $1 < \phi < -1$, the effect of each compounds over time and the process explodes; if $\phi = \pm 1$, the shocks and the initial condition have a permanent effect of the process.

Now, instead of having y_t depend on only its previous value and a contemporary shock, we'll have it depend on the previous p values

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + w_t.$$

It is often more convenient to transform this into a first-order vector-valued equation,

$$\boldsymbol{\xi}_t = \boldsymbol{F}\boldsymbol{\xi}_{t-1} + \boldsymbol{v}_t \tag{5}$$

where

$$\boldsymbol{\xi}_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}$$

is $p \times 1$,

$$\boldsymbol{F} = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

is $p \times p$, and

$$\boldsymbol{v}_t = \begin{pmatrix} w_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

By a similar recursive operation before, if we have an initial condition at ξ_{t-1} we can solve for the state of the process at some future horizon j,

$$
\begin{aligned}
\boldsymbol{\xi}_t &= \boldsymbol{F}\boldsymbol{\xi}_{t-1} + \boldsymbol{v}_t, \\
\boldsymbol{\xi}_{t+1} &= \boldsymbol{F}\boldsymbol{\xi}_t + \boldsymbol{v}_t && = \boldsymbol{F}(\boldsymbol{F}\boldsymbol{\xi}_{t-1} + \boldsymbol{v}_t) + \boldsymbol{v}_{t+1} && = \boldsymbol{F}^2\boldsymbol{\xi}_{t-1} + \boldsymbol{F}\boldsymbol{v}_t + \boldsymbol{v}_{t+1}, \\
\boldsymbol{\xi}_{t+2} &= \boldsymbol{F}\boldsymbol{\xi}_{t+1} + \boldsymbol{v}_{t+2} && = \boldsymbol{F}(\boldsymbol{F}^2\boldsymbol{\xi}_{t-1} + \boldsymbol{F}\boldsymbol{v}_t + \boldsymbol{v}_{t+1}) + \boldsymbol{v}_{t+2} && = \boldsymbol{F}^3\boldsymbol{\xi}_{t-1} + \boldsymbol{F}^2\boldsymbol{v}_t + \boldsymbol{F}\boldsymbol{v}_{t+1} + \boldsymbol{v}_{t+2}, \\
&\vdots \\
\boldsymbol{\xi}_{t+j} &= \boldsymbol{F}^{j+1}\boldsymbol{\xi}_{t-1} + \sum_{k=0}^{j} \boldsymbol{F}^{j-k}\boldsymbol{v}_{t-k} && && (6)
\end{aligned}
$$

Note that the future value of interest y_{t+j} is the first row of this last equation. If f_{ij}^n is the (i,j) element of F^n, then

$$
\begin{aligned}
y_{t+j} = & \underbrace{f_{11}^{j+1} y_{t-1} + f_{12}^{j+1} y_{t-2} + \cdots + f_{1p}^{j+1} y_{t-p}}_{\text{initial conditions}} \\
& \underbrace{+ w_{t+j} + f_{11} w_{t+j-1} + \cdots + f_{11}^{j} w_t}_{\text{effect of shocks since initial state}}
\end{aligned}
$$

We can start to see how a process with a complicated structure can have its future values predicted, and how complicated the relationship between past and future values can be. For example, note how the effects of the shocks and the most recent period are all mediated by the $(1,1)$ element of F^n.

$$
\begin{aligned}
f_{11}^1 &= \phi_1 \\
f_{11}^2 = (\boldsymbol{F}\boldsymbol{F})_{1,1} &= \left(\begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} \phi_1 & \cdots \\ 1 & \cdots \\ 0 & \cdots \\ \vdots & \cdots \\ 0 & \cdots \end{pmatrix} \right)_{1,1} = \phi_1^2 + \phi_2 \\
f_{11}^3 = (\boldsymbol{F}^2\boldsymbol{F})_{1,1} &= \left(\begin{pmatrix} \phi_1^2 + \phi_2 & \phi_1\phi_2 + \phi_3 & \cdots & \phi_1\phi_{p-1} + \phi_p & \phi_1\phi_p \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} \phi_1 & \cdots \\ 1 & \cdots \\ 0 & \cdots \\ \vdots & \cdots \\ 0 & \cdots \end{pmatrix} \right)_{1,1} = \phi_1^3 + 2\phi_1\phi_2 + \phi_3 \\
&\vdots
\end{aligned}
$$

It is easy to see how the number of terms of f_{11}^n blows up as n increases. This is why it is often easier to reason with the vector-valued form of the system as shown in Equation 6. If we want to get any kind of intuition for how the previous value we observe

of a process may influence future values, it is worthwhile to work more with this vector-valued representation because it can provide analytical solutions for these influences.

If all of the eigenvalues of $\boldsymbol{F}$ are distinct, then we can write

$$\boldsymbol{F} = \boldsymbol{T}\boldsymbol{\Lambda}\boldsymbol{T}^{-1} \leftrightarrow \boldsymbol{F}^j = \boldsymbol{T}\boldsymbol{\Lambda}^j\boldsymbol{T}^{-1}$$

where $\boldsymbol{\Lambda}$ is a diagonal matrix of eigenvalues of $\boldsymbol{F}$. After some algebraic manipulations, we find that

$$\frac{\partial y_{t+j}}{\partial w_t} = c_1\lambda_1^j + \cdots + c_p\lambda_p^j$$

where λ_k are the eigenvalues of $\boldsymbol{F}$ and $c_1 + \cdots + c_p = 1$. We can see that a similar exponential decay in the effect of the shock w_t occurs for a pth-order difference equations as for a first-order. Multiple lags do not get around the linearity of the system, which always leads to the effect of a shock either growing or decaying exponentially and monotonically over time.

Lag Operators

This chapter is essentially a distillation of Hamilton's [2] chapter of the same name.

[2] James D Hamilton. *Time series analysis*, volume 2. Princeton Princeton University Press, 1994

The lag operator L is a convenient notational tool that allows difference equations and their properties to be expressed in elegant ways. For example, it allows us to write a first-order difference equation in the following equivalent ways,

$$
\begin{aligned}
y_t &= \phi y_{t-1} + w_t \\
&\updownarrow \\
y_t &= \phi L y_t + w_t \\
&\updownarrow \\
(1-\phi L) y_t &= w_t \\
&\updownarrow \\
(1+\phi L+\phi^2 L^2+\cdots+\phi^t L^t)(1-\phi L) y_t &= (1+\phi L+\phi^2 L^2+\cdots+\phi^t L^t) w_t \\
&\updownarrow \\
(1-\phi^{t+1} L^{t+1}) y_t &= (1+\phi L+\phi^2 L^2+\cdots+\phi^t L^t) w_t
\end{aligned}
$$

There are two things to note about the last line here. First, that it is equivalent to Equation 2 with $t = 0$ and $j = t$. A second, if $|\phi| < 1$, as t grows large, $(1-\phi^{t+1} L^{t+1}) y_t \approx y_t$. So, we can define a new operator

$$
(1-\phi L)^{-1} = \lim_{t\to\infty} (1+\phi L+\cdots+\phi^t L^t)
$$

and write

$$
(1-\phi L) y_t = w_t \leftrightarrow y_t = (1-\phi L)^{-1} w_t
$$

Note that it is only possible to express the process this way if $|\phi| < 1$.

If we have a p-th order difference equation, we can write

$$
\begin{aligned}
y_t &= \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + w_t \\
&\updownarrow \\
(1 - \phi_1 L - \cdots - \phi_p L^p) y_t &= w_t \qquad \updownarrow \\
(1 - \lambda_1 L) * \cdots * (1 - \lambda_p L) y_t &= w_t
\end{aligned}
$$

Finding the λ_k values is similar to factoring any other polynomial, and in fact it is the identical operation of find the eigenvalues of $\boldsymbol{F}$ in Equation 6. Given this symmetry, if all of the eigenvalues found lie within the unit circle, then the system is stable, and we can write

$$y_t = (1 - \lambda_1 L)^{-1} * \cdots * (1 - \lambda_p L)^{-1} w_t$$

After some involved algebra, we also find, unsurprisingly, that the effect of a shock w_t at some horizon j is the same as well.

Stationary ARMA Processes

This chapter is essentially a distillation of Hamilton's [3] chapter of the same name.

[3] James D Hamilton. *Time series analysis*, volume 2. Princeton Princeton University Press, 1994

A process Y_t is **weakly stationary** if

$$\begin{aligned} \mathbb{E}(Y_t) &= \mu & \forall t \\ \mathbb{E}[(Y_t - \mu)(Y_{t-j} - \mu)] &= \gamma_j & \forall t, j \end{aligned}$$

A process Y_t is **strongly stationary** if for any values $j_1, ..., j_n$ the joint distribution of $(Y_t, Y_{t+j_1}, ..., Y_{t+j_n})$ only depends on the dates separating $j_1, ..., j_n$ and not on t.

A process Y_t is **ergodic** for a certain characteristic (its mean, second moment, etc) if observing that time series over a long period of time allows you to make claims about that characteristic at a point in time. For example, for a time series, we never actually observe the processes repeatedly at any specific time. This would normally make it impossible to say what its mean is at any point in time. If the time series was ergodic for the mean, then we can say that a mean estimated from sufficient observations of a single time series across time is also a good estimate of the mean at any point in time.

Notes to include

state space models - local level model

two main assumptions are mentioned: the authors will assume normality or, when normality is not assumed, minimum variance linear unbiased estimation. I'm assuming the latter is BLUE. I have never thought that assuming normality and assuming BLUE were somehow solutions to the same problem. That's interesting.

A basic general time series model can be represented by

$$y_t = \mu_t + \gamma_t + \epsilon_t$$

- $y_t \in \{y_1, ..., y_n\}$, the observations
- μ_t, trend (slowly varying, the state)
- γ_t, seasonal (periodic)
- ϵ_t, error disturbance

In state-space time series estimation, we use the observations to make inferences about the state.

In general, when creating a state space model, we think about how to model μ_t and γ_t. A more specific, but still applicable model that illustrates this is the following random walk model known as a "local level model"

$$\begin{aligned} y_t &= \alpha_t + \epsilon_t, & \epsilon_t &\sim N(0, \sigma_\epsilon^2), \\ \alpha_{t+1} &= \alpha_t + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2), \\ & & \epsilon_t &\perp \eta_t. \end{aligned}$$

Here, the state is α_t, and there is no seasonal component. The error component ϵ_t of y_t differs from other representations of random walks in that it has a purely contemporaneous effect; it does not accumulate over time, the shocks η_t to the state α_t do. It is as if we have a random walk with an extra disturnace term.

Additionally, we also assume $\alpha_1 \sim N(a_1, P_1)$, and that $a_1, P_1, \sigma_\epsilon^2, \sigma_\eta^2$ are known.

Since we are assuming a normal distribution for our fundamental variables and paramters, would could obtain closed form solutions for the moments of y_t at any time t, but this is intractable as n grows. This is why we use the technique known as filtering.

filtering

Assume

- $Y_{t-1} = (y_1, ..., y_{t-1})'$ for $t = 2, 3, ..., n$.
- $\alpha_t | Y_{t-1} \sim N(a_t, P_t)$, where a_t, P_t are known.
- $\alpha_t | Y_t \sim N(a_{t|t}, P_{t|t})$.
- $\alpha_{t+1} | Y_t \sim N(a_{t+1}, P_{t+1})$.

The first thing that jumps out is that α_t is always unknown. We make guesses about them by estimating a_t and P_t. We fill focus on two estimates involving the state of the system, a one-step ahead forecast a_{t+1} and a "filtered-estimator" $a_{t|t}$ of the current state α_t. Both estimates are made using information at time t, Y_t.

Filtering makes use of the one-step ahead prediction error of y_t, $v_t = y_t - a_t$. Some important moments of this prediction error are

$$\begin{aligned}
\mathbb{E}(v_t | Y_{t-1}) &= \mathbb{E}(\alpha_t + \epsilon_t - a_t | Y_{t-1}) = a_t - a_t = 0, \\
Var(v_t | Y_{t-1}) &= Var(\alpha_t + \epsilon_t - a_t | Y_{t-1}) = P_t + \sigma_\epsilon^2, \\
\mathbb{E}(v_t | \alpha_t, Y_{t-1}) &= \mathbb{E}(\alpha_t + \epsilon_t - a_t | \alpha_t, Y_{t-1}) = \alpha_t - a_t, \\
Var(v_t | \alpha_t, Y_{t-1}) &= Var(\alpha_t + \epsilon_t - a_t | \alpha_t, Y_{t-1}) = \sigma_\epsilon^2,
\end{aligned}$$

Remember that at $t-1$, a_t and P_t are known, so $\mathbb{E}(a_t | Y_{t-1}) = a_t$ and $Var(a_t | Y_{t-1}) = 0$. We can update our expectations of the state with our prediction error by noticing that $p(\alpha_t | Y_t) = p(\alpha_t | Y_{t-1}, v_t)$ because if Y_t is fixed, then so are both (Y_{t-1}, y_t) and (Y_{t-1}, v_t) because a_t is known at time t and $v_t = y_t - a_t$. Remembering that $p(\cdot)$ is a density function and all of our densities in this example are normal,

$$\begin{aligned}
p(\alpha_t | Y_{t-1}, v_t) &= p(\alpha_t, v_t | Y_{t-1}) / p(v_t | Y_{t-1}), \\
&= p(\alpha_t | Y_{t-1}) p(v_t | \alpha_t, Y_{t-1}) / p(v_t | Y_{t-1}), \\
&= \text{constant} \times \exp\left(\frac{-1}{2} Q \right).
\end{aligned}$$

And remember that the normal density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(\frac{-1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$$

so whatever Q is informs us of μ and σ for the resulting normal distribution of interest, and thus defines it. Q is built from the moments above

$$\begin{aligned} Q &= \frac{(\alpha_t - a_t)^2}{P_t} + \frac{(v_t - \alpha_t + a_t)^2}{\sigma_\epsilon^2} - \frac{v_t^2}{(P_t + \sigma_\epsilon^2)} \\ &= \frac{\alpha_t - \left(a_t + \frac{P_t v_t}{P_t + \sigma_\epsilon^2}\right)}{\frac{P_t \sigma_\epsilon^2}{P_t + \sigma_\epsilon^2}} \end{aligned}$$

so

$$p(\alpha_t|Y_{t-1}, v_t) = p(\alpha_t|Y_t) = N\left(a_t + \frac{P_t v_t}{P_t + \sigma_\epsilon^2}, \frac{P_t \sigma_\epsilon^2}{P_t + \sigma_\epsilon^2}\right)$$

and we defined $p(\alpha_t|Y_t) = N(a_{t|t}, P_{t|t})$ so

$$\begin{aligned} a_{t|t} &= a_t + \frac{P_t v_t}{P_t + \sigma_\epsilon^2}, \\ P_{t|t} &= \frac{P_t \sigma_\epsilon^2}{P_t + \sigma_\epsilon^2}, \end{aligned}$$

are respectively the filter estimate of the current state α_t, which is the most up-to-date expected value of the current state, and our most up-to-date quantification of how far away from this expectation the actual state may be (i.e. the most up to date variance of the state).

This up-to-date distribution of the current state is different from the unupdated verion, $p(\alpha_t|Y_{t-1}) = N(a_t, P_t)$. To understand this difference, we first need to understand the term $P_t + \sigma_\epsilon^2$. This is the total variance of our model given the information we have at time $t-1$, which is the same as the variance of our prediction error conditional on that same information,

$$Var(y_t|Y_{t-1}) = Var(v_t|Y_{t-1}) = Var(\alpha_t + \epsilon_t|Y_{t-1}) = P_t + \sigma_\epsilon^2.$$

Thus

$$\frac{P_t}{P_t + \sigma_\epsilon^2}$$

represents how much of our prediction error is likely due to changes in the underlying state. This is the scale factor on the prediction error. It says that because a part of every observation is due to noise,

we don't take every prediciton error 100% seriously. Instead, we adjust our expectation of the state according to how much we think a prediction deviated from an observation because the state unexpectedly changed, rather than because contemporaneous noise was especially high. How much we anticipate state to unexpectedly change is P_t, and we update scale by

$$\frac{\sigma_\epsilon^2}{P_t + \sigma_\epsilon^2},$$

the share of noise in prediction error variance. Note that this always increases our certainty about the state, which makes sense given we are adding new information to the model. We may not increase our certainty very much, however, if our observations and hence prediction errors are very noisey.

P_t is calculated with each observation as a part of the one-step ahead forecast process, and σ_ϵ^2 is assumed to be known. This forecast process appears to be a part of both fitting the model as well as forecasting itself.

To make our one-step ahead forecast for the state of our system, we go back to our state model,

$$a_{t+1} = \mathbb{E}(\alpha_t + \eta_t | Y_t) = \mathbb{E}(\alpha_t | Y_t) = a_{t|t} = a_t + \frac{P_t v_t}{P_t + \sigma_\epsilon^2}$$

and

$$P_{t+1} = Var(\alpha_t + \eta_t | Y_t) = P_{t|t} + \sigma_\eta^2 = \frac{P_t \sigma_\epsilon^2}{P_t + \sigma_\epsilon^2} + \sigma_\eta^2.$$

Because we have a random walk, our best forecast of the state's mean is our most up-to-date expectation, and the variance of that forecast is our most up-to-date variance of the state plus our current understanding of how that state's random walk evolves.

Perhaps unsurprisingly, one of the scaling factors introduced above plays a special role in filtering,

$$F_t = Var(v_t | Y_{t-1}) = P_t + \sigma_\epsilon^2, \qquad K_t = \frac{P_t}{F_t}.$$

The **Kalman gain** K_t is the scaling factor on the prediction errors. We can succintly write the rules for updating from t to $t+1$ as

$$\begin{aligned} v_t &= y_t - a_t, & F_t &= P_t + \sigma_\epsilon^2, \\ a_{t|t} &= a_t + K_t v_t, & P_{t|t} &= P_t(1 - K_t), \\ a_{t+1} &= a_t + K_t v_t, & P_{t+1} &= P_t(1 - K_t) + \sigma_\eta^2. \end{aligned}$$

Forecast errors

The forecast errors v_t and their variance F_t are used in a variety of ways in state space modeling because they represent the new information we learn about y_t that could not be predicted given our past observations. The forecast errors are often referred to as innovations because they represent new information in this way.

The forecast errors are mutually independent. To see this, start with the joint probability of a set of observations and use the definition of a marginal distribution

$$p(y_1, ..., y_n) = p(y_1) \prod_{t=2}^{n} p(y_t | Y_{t-1}).$$

To understand what happens from here, we need to understand how to apply a change of variables.

Aside: Change of Variable

Let's start with the one dimensional case. If we have a random variable X with density function $f_X(x)$ and we wish to find the density of another random variable $Y = g(X)$, denoted $f_Y(y)$, and if g is strictly monotonic, then we can find it with

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

The absolute value is a scaling factor derived from the fact that, given the relationship $Y = g(X)$, the probability that Y lies in a certain image must equal the probability that X lies in the associated pre-image (note the proof of the transition from the first to second line requires knowledge of Real Analysis):

$$\begin{aligned}
\mathbb{P}[Y \in [y, y + dy]] &= \mathbb{P}[X \in [x, x + dx]] \\
f_Y(y)|dy| &= f_X(x)|dx| \\
f_Y(y) &= f_X(x) \left| \frac{dx}{dy} \right| \\
f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|
\end{aligned}$$

In the multivariate (i.e. vector) case we have

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) \left| \det(J_{g^{-1}}(\boldsymbol{y})) \right|$$

Let $v_t = g(y_t) = y_t - a_t$ so $y_t = g^{-1}(v_t) = v_t + a_t$ and $\frac{dg^{-1}(v_t)}{dv_t} = 1$, and letting $\boldsymbol{y}_t$ and $\boldsymbol{v}_t$ denote the random variables of which y_t and v_t are observations, we can then write

$$\begin{aligned} p(y_1, ..., y_n) &= f_{\boldsymbol{y}_1}(y_1) \prod_{t=2}^{n} f_{\boldsymbol{y}_t}(y_t | Y_{t-1}) \\ &= f_{\boldsymbol{y}_1}(v_1 + a_1) \prod_{t=2}^{n} f_{\boldsymbol{y}_t}(v_t + a_t | Y_{t-1}) \\ &= f_{\boldsymbol{v}_1}(v_1) \prod_{t=2}^{n} f_{\boldsymbol{v}_t}(v_t | Y_{t-1}) \end{aligned}$$

The second equality holds simply because $y_t = v_t + a_t$; we have not truly changed variables yet because the underlying distribution is still that of y_t. In the third equality, we change to the variable to v_t. This has an especially simply form

$$\begin{aligned} f_{\boldsymbol{v}_1}(v_1) &= f_{\boldsymbol{y}_1}(v_1 + a_1), \\ f_{\boldsymbol{v}_t | \boldsymbol{Y}_{t-1}}(v_t \mid Y_{t-1}) &= f_{\boldsymbol{y}_t | \boldsymbol{Y}_{t-1}}(v_t + a_t \mid Y_{t-1}) \text{ when } t > 1, \end{aligned}$$

because y_t is just a shift of v_t, and the shift a_t is known conditional on Y_{t-1}. It may seem weird to claim that two distributions that are shifts of eachother are equal. Bear in mind, however, that the functions in the equations above have become functions of v_t, so it is not that we are somehow claiming that $f_{\boldsymbol{v}_t \sim \boldsymbol{Y}_{t-1}}$ perfetly overlaps $f_{\boldsymbol{y}_t \sim \boldsymbol{Y}_{t-1}}$; we are saying that the probability density of $\boldsymbol{v}_t$ taking on a certain value can be equivalently expressed in terms of either $f_{\boldsymbol{v}_t \sim \boldsymbol{Y}_{t-1}}$ or $f_{\boldsymbol{y}_t \sim \boldsymbol{Y}_{t-1}}$ when the input to the latter is shifted appropriately; this is simply the core result of the change of variable theorem, applied here.

The last thing to do is to show that

$$f_{\boldsymbol{v}_t | \boldsymbol{Y}_{t-1}}(v_t \mid Y_{t-1}) = f_{\boldsymbol{v}_t}(v_t)$$

We already know that $v_t \sim N(0, F_t)$, so any conditional dependence on Y_{t-1} must come through F_t. But

$$F_t = P_t + \sigma_\epsilon^2$$

and

$$P_{t+1} = \frac{P_t \sigma_\epsilon^2}{P_t + \sigma_\epsilon^2} + \sigma_\eta^2$$

where σ_ϵ^2, σ_η^2, and P_1 are known, so the evolution of F_t is *deterministic* because the evolution of P_t is deterministic. A deterministic process does not depend on any random variable, so

$$f_{\boldsymbol{v}_t | \boldsymbol{Y}_{t-1}}(v_t \mid Y_{t-1}) = f_{\boldsymbol{v}_t}(v_t).$$

This means that we can finally write

$$\begin{aligned} p(y_1, ..., y_n) &= f_{v_1}(v_1) \prod_{t=2}^{n} f_{v_t}(v_t | Y_{t-1}) \\ &= \prod_{t=1}^{n} f_{v_t}(v_t) \\ &= p(v_1, ..., v_n) \end{aligned}$$

and we have shown that the prediction errors are mutually independent (we could technically write that last line earlier, but I wrote it here to cement the point).

We now show that the forecast errors can be computed by a Cholesky decomposition. To recap, our updating rules are

$$\begin{aligned} v_t &= y_t - a_t, & F_t &= P_t + \sigma_\epsilon^2, \\ a_{t|t} &= a_t + K_t v_t, & P_{t|t} &= P_t(1 - K_t), \\ a_{t+1} &= a_t + K_t v_t, & P_{t+1} &= P_t(1 - K_t) + \sigma_\eta^2, \\ & & K_t &= \frac{P_t}{F_t}. \end{aligned}$$

We can recursively solve for the forecast errors given a_1 and the observations Y_n:

$$\begin{aligned} v_1 &= y_1 - a_1 \\ v_2 = y_2 - a_2 = y_2 - (a_1 + K_1 v_1) &= y_2 - a_1 - K_1(y_1 - a_1) \\ v_3 = y_3 - a_3 = y_3 - (a_2 + K_2 v_2) &= y_3 - a_2 - K_2(y_2 - a_1 - K_1(y_1 - a_1)) \\ &= y_3 - (a_1 + K_1 v_1) - K_2(y_2 - a_1) + K_2 K_1(y_1 - a_1) \\ &= y_3 - a_1 - K_1(y_1 - a_1) - K_2(y_2 - a_1) + K_2 K_1(y_1 - a_1) \\ &= y_3 - a_1 - K_2(y_2 - a_1) - K_1(y_1 - a_1) + K_2 K_1(y_1 - a_1) \\ &= y_3 - a_1 - K_2(y_2 - a_1) - K_1(y_1 - a_1)(1 - K_2) \\ &= y_3 - a_1 - K_2(y_2 - a_1) - K_1(1 - K_2)(y_1 - a_1) \end{aligned}$$

We can represent this in matrix form,

$$v = C(Y_n - 1a_1), \text{ with } v = (v_1 \cdots v_n)'$$

The critical part of the above is the $n \times n$ matrix C, whose structure we inductively describe above,

$$C = \begin{pmatrix} 1 & 0 & 0 & & 0 \\ c_{21} & 1 & 0 & & 0 \\ c_{31} & c_{32} & 1 & & 0 \\ & & & \ddots & \vdots \\ c_{n1} & c_{n2} & c_{n3} & \cdots & 1 \end{pmatrix},$$

$$\left.\begin{aligned} c_{i,i-1} &= -K_{i-1}, \\ c_{ij} &= -(1-K_{i-1})(1-K_{i-2})\cdots(1-K_{j+1})K_j \end{aligned}\right\} \quad \begin{aligned} &i > j \\ &\text{for } i = 2, ..., n \\ &j = 1, ..., i-2 \end{aligned}$$

The indexes of this last line are a bit hard to understand, so let's apply them to the closed form of v_3 we derived above,

$$v_3 = y_3 - a_1 - K_2(y_2 - a_1) - \underbrace{K_1(1-K_2)(y_1 - a_1)}_{c_{31}}.$$

Since we are always in the lower triangular half of the matrix, we essentially create the coefficient on $y_1 - a_1$ by factors from right to left, "counting" from j to $i-1$. We can then state

$$v \sim N(0, C\Omega C')$$

where $\Omega = \text{Var}(Y_n)$ as define by the autocovariance structure of the local level model we have been working with.

Now, we also know that $\text{Var}(v_t) = F_t$ and $\text{Cov}(v_t, v_s) = 0$ for $s \neq t$, so we can also write

$$v \sim N(0, F)$$

where F is a diagonal matrix whose non-zero elements are $F_1, ..., F_n$. So, we can say

$$F = C\Omega C',$$

which is exactly the form of a Cholesky decomposition: the representation of a positive semidefinite matrix Ω as a diagonal matrix F. It is also userful to notice that $\Omega^{-1} = C'F^{-1}C$, which should be familiar with Cholesky decomposition for OLS estimation.

Next we define the state estimation error,

$$x_t = \alpha_t - a_t.$$

Apparently $\text{Var}(x_t) = P_t$, but I don't see how this is is true unconditionally, so let's try and derive it.

$$\begin{aligned}
\text{Var}(x_t) &= \text{Var}(\alpha_t - a_t) = \text{Var}(\alpha_t) + \text{Var}(a_t) \\
&= \mathbb{E}(\alpha_t^2) - \mathbb{E}(\alpha_t)^2 + \mathbb{E}(a_t^2) - \mathbb{E}(a_t)^2 \\
&= \mathbb{E}(\mathbb{E}(\alpha_t^2 \mid Y_{t-1})) - \mathbb{E}(\mathbb{E}(\alpha_t \mid Y_{t-1}))^2 + \mathbb{E}(\mathbb{E}(a_t^2 \mid Y_{t-1})) - \mathbb{E}(\mathbb{E}(a_t \mid Y_{t-1}))^2 \\
&= \mathbb{E}(\mathbb{E}(\alpha_t^2 \mid Y_{t-1}) - \mathbb{E}(\alpha_t \mid Y_{t-1})^2) \\
&= \mathbb{E}(\text{Var}(\alpha_t \mid Y_{t-1})) \\
&= \mathbb{E}(P_t) \\
&= P_t
\end{aligned}$$

Again, the last line holds because P_t is a deterministic process given that P_1, σ_ϵ^2, and σ_η^2 are known. The fourth line holds because as of $t-1$, a_t is known. The use of the law of iterated expectations seems a little bit like magic: how can we go from being interested in the unconditional variance of our state α_t and our state forecast a_t to being satisfied with a statement that depends on knowledge as of $t-1$? Unconditionally, our state and forecast could be *anything*, right? Well, I think the answer comes down to the fact that P_t holds a lot of information. The error variances σ_ϵ^2 and σ_η^2 that dictate its evolution are known *because* we know what our model is and observations are (I'm still now sure how these are known given those, but that is what the book states). So, while P_t is defined conditionally ($\alpha_t \mid Y_{t-1} \sim N(a_t, P_t)$), its true value is *not conditional*. This is why we can know what the unconditional variation in α_t and x_t by conditioning on Y_{t-1} in an application of the law of iterated expectations.

Given the definition of x_t, we can express our forecast error in terms of our state estimation error and state update rules for our errors between t and $t+1$ that are analogous to the update rules for our model,

$$\begin{aligned}
v_t &= y_t - a_t = \alpha_t + \epsilon_t - a_t = x_t + \epsilon_t \\
x_{t+1} &= \alpha_{t+1} - a_{t+1} \\
&= \alpha_t + \eta_t - a_t - K_t v_t \\
&= x_t + \eta_t - K_t(x_t + \epsilon_t) \\
&= L_t x_t + \eta_t - K_t \epsilon_t.
\end{aligned}$$

where $L_t = 1 - K_t$. So, our state error next period is the state error of this period attenuated by the share of the variance of noise in our observations in total variance of our observations, $1 - K_t = \sigma_\epsilon^2/(P_t + \sigma_\epsilon^2)$, adjusting for the random walk of our state and the noise in our observation this period. Note that the random walk of our state directly affects state esetimation error next period, becauase it is unpredictable, while the noise in the observation this period

affects it via the Kalman gain, K_t; this is because the Kalman gain describes how much of the observed variance is due to variation in our state as opposed to noise in the observations, so said noise's affect on state estimation error is reduced in proportion to it: only part of the noise this period is allowed to affect our state estimation, so only a part of it should affect our state estimation error.

State Smoothing

So, how do we estimate $\alpha_1, ..., \alpha_n$ given our sample Y_n? We can state the following

$$\begin{aligned} \alpha_t \mid Y_n &\sim N(\hat{\alpha}_t, V_t), \\ \hat{\alpha}_t &= \mathbb{E}(\alpha_t \mid Y_n), && \text{the smoothed state,} \\ V_t &= \text{Var}(\alpha_t \mid Y_n), && \text{smoothed state variance.} \end{aligned}$$

Also, remember the following results proved above

1. All forecast errors $v_1, ..., v_n$ are mutually independent.
2. The "future" forecast errors $v_t, ..., v_n$ are independent of past observations $y_1, ..., y_{t-1}$ and have zero mean.
3. Y_n is fixed if and only if Y_{t-1} and $v_t, ..., v_n$ are fixed.

Then also we know in the bivariate normal case, we can state for two such variables x and y

$$\mathbb{E}(x \mid y) = \mu_x + \frac{\sigma_{xy}}{\sigma_y^2}(y - \mu_y), \qquad \text{Var}(x \mid y) = \sigma_x^2 + \frac{\sigma_{xy}^2}{\sigma_y^2},$$

where $\frac{\sigma_{xy}}{\sigma_y^2}$ is the theoretical value of the OLS coefficient of x on y. If we note that

$$\hat{\alpha}_t = \mathbb{E}(\alpha_t \mid Y_n)$$

then we can extend this to the multivariate case and apply it to our local level model:

$$\begin{aligned} p(\alpha_t, Y_n) &= p(Y_n)p(\alpha_t \mid Y_n) \\ f_{\alpha_t, Y_n}(\alpha_t, Y_n) &= f_{Y_n}(Y_n) f_{\alpha_t \mid Y_n}(\alpha_t \mid Y_n) \\ \frac{f_{\alpha_t, Y_n}(\alpha_t, Y_n)}{f_{Y_n}(Y_n)} &= f_{\alpha_t \mid Y_n}(\alpha_t \mid Y_n) \end{aligned}$$

and we have for our local level model

$$Y_n \sim N(\mathbf{1}a_1, \Omega)$$
$$\alpha_t = y_t - \epsilon_t \sim N(a_1, \sigma_\epsilon^2 + (t-1)\sigma_\eta^2 + \sigma_\epsilon^2)$$

To understand where this is coming from, note that our model of y_t can be expressed as

$$y_t = a_1 + \sum_{j=1}^{t-1} \eta_j + \epsilon_t$$

because each period we have ϵ_t and η_t but η_t only affects the next period, more explicitly

$$Y_n = \begin{pmatrix} a_1 + \epsilon_1 \\ a_1 + \eta_1 + \epsilon_2 \\ a_1 + \eta_1 + \eta_2 + \epsilon_3 \\ \vdots \\ a_1 + \sum_{t=1}^{n-1} \eta_t + \epsilon_n \end{pmatrix}$$

and so

$$\begin{aligned} \Omega = \text{Var}(Y_n) &= \mathbb{E}[(Y_n - \mathbb{E}(Y_n))(Y_n - \mathbb{E})'] \\ &= \begin{pmatrix} \text{Var}(y_1) & \text{Cov}(y_1, y_2) & \cdots & \text{Cov}(y_1, y_n) \\ \text{Cov}(y_2, y_1) & \text{Var}(y_2) & & \\ \vdots & & \ddots & \\ \text{Cov}(y_n, y_1) & & & \text{Var}(y_n) \end{pmatrix} \\ &= \begin{pmatrix} P_1 + \sigma_\epsilon^2 & P_1 & P_1 & \cdots & P_1 \\ P_1 & P_1 + \sigma_\eta^2 + \sigma_\epsilon^2 & P_1 + \sigma_\eta^2 & & \\ P_1 & P_1 + \sigma_\eta^2 & P_1 + 2\sigma_\eta^2 + \sigma_\epsilon^2 & & \\ \vdots & & & \ddots & \\ P_1 & & & & P_1 + (n-1)\sigma_\eta^2 + \sigma_\epsilon^2 \end{pmatrix} \end{aligned}$$

Note how the variances due to shocks to the random walk enter the elements and where they are absent. They enter when two observations share historical shocks: σ_η^2 is present after the first period, in proportion to the overlap in historical periods (i.e. in proportion to the earlier observations shocks). Also note where variance to the idiosyncratic shocks ϵ_t appears: only on diagonal elements.

$$\begin{pmatrix} \alpha_t \\ Y_n \end{pmatrix} \sim N\left(a_t \mathbf{1}_{n+1}, \begin{pmatrix} \text{Var}(\alpha_t) & \text{Cov}(\alpha_t, Y_n') \\ \text{Cov}(Y_n, \alpha_t) & \text{Var}(Y_n) \end{pmatrix} \right)$$

where

$$\begin{pmatrix} \text{Var}(\alpha_t) & \text{Cov}(\alpha_t, Y_n') \\ \text{Cov}(Y_n, \alpha_t) & \text{Var}(Y_n) \end{pmatrix} = \begin{pmatrix} \text{Var}(y_t - \epsilon_t) & \text{Cov}(y_t - \epsilon_t, y_1) & \cdots & \text{Cov}(y_t - \epsilon_t, y_t) & \cdots & \text{Cov}(y_t - \epsilon_t, y_n) \\ \text{Cov}(y_1, y_t - \epsilon_t) & & & & & \\ \vdots & & & & & \\ \text{Cov}(y_t, y_t - \epsilon_t) & & & \Omega & & \\ \vdots & & & & & \\ \text{Cov}(y_n, y_t - \epsilon_t) & & & & & \end{pmatrix}$$

We are not adding redundant data to the vector Y_n because we are adding the state explicitly, which requires considering the difference between the observation and its error as opposed to just the observation, therefore adding new information to the system. The multivariate normal distribution is

$$f_X(x_1, ..., x_n) = \frac{1}{\sqrt{(2\pi)^n |\mathbf{\Sigma}|}} \exp\left(\frac{-1}{2} (\boldsymbol{x} - \boldsymbol{\mu}_x)' \mathbf{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_x) \right) = N(\boldsymbol{\mu}_x, \mathbf{\Sigma})$$

so

$$f_{\alpha_t, Y_n}(\alpha_t, Y_n) = \text{const.} \times \exp\left(\frac{-1}{2} \left(\begin{pmatrix} \alpha_t \\ Y_n \end{pmatrix} - a_1 \mathbf{1}_{n+1} \right)' \begin{pmatrix} \text{Var}(\alpha_t) & \text{Cov}(\alpha_t, Y_n') \\ \text{Cov}(Y_n, \alpha_t) & \Omega \end{pmatrix}^{-1} \left(\begin{pmatrix} \alpha_t \\ Y_n \end{pmatrix} - a_1 \mathbf{1}_{n+1} \right) \right)$$

and

$$f_{Y_n}(Y_n) = \text{const.} \times \exp\left(\frac{-1}{2} (Y_n - a_1 \mathbf{1}_n)' \Omega^{-1} (Y_n - a_1 \mathbf{1}_n) \right)$$

All the action is taking place in the exponential, so, dropping the $-1/2$,

$$\begin{aligned} \text{exponential} = & \left(\begin{pmatrix} \alpha_t \\ Y_n \end{pmatrix} - a_1 \mathbf{1}_{n+1} \right)' \begin{pmatrix} \text{Var}(\alpha_t) & \text{Cov}(\alpha_t, Y_n') \\ \text{Cov}(Y_n, \alpha_t) & \Omega \end{pmatrix}^{-1} \left(\begin{pmatrix} \alpha_t \\ Y_n \end{pmatrix} - a_1 \mathbf{1}_{n+1} \right) \\ & - (Y_n - a_1 \mathbf{1}_n)' \Omega^{-1} (Y_n - a_1 \mathbf{1}_n) \end{aligned}$$

We want to isolate what equates to the mean of α_t. Consequently, we don't really care about the second term being substracted, because it will only affect terms in the final expression that haveelements of Y_n, not α_t.

Bibliography

[1] James D Hamilton. *Time series analysis*, volume 2. Princeton Princeton University Press, 1994.